

4.3 Converting with Base Systems

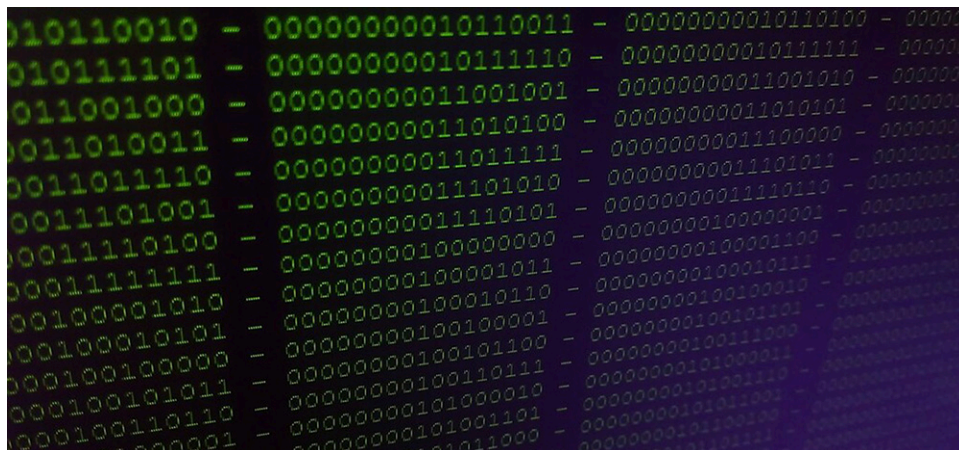


Figure 4.4 Computers use Base 2, which only uses 0's and 1's, to represent quantity. (credit: modification of work "IMAG0933" by yvanhou/Flickr, CC BY 2.0)

Learning Objectives

After completing this section, you should be able to:

1. Convert another base to base 10.
2. Write numbers in different base systems.
3. Convert base 10 to other bases.
4. Determine errors in converting between bases.

In our system of numbers, we use base 10, but using base 10 was not a given within other systems. There were other systems that used bases other than 10, as we saw with the Mayans and the Babylonians. The base 10 system comes down to grouping objects in sets of 10, but grouping in sets of 10 only happens if the culture values grouping by that many. We feel 10 is natural because we have 10 fingers. There are other systems using other grouping values, such as 4 or 20.

One good reason for examining other bases is to remind ourselves how we had to learn arithmetic when we were young, memorizing rules for our base 10 system. We had to learn why those arithmetic rules made sense, such as why $1 + 1 = 2$ and $1 + 2 = 3$. Another good reason for learning other base systems is due to computers; their circuitry instead uses base 2.

In this section, we explore other base systems and how to convert between them.

Conversion of Another Base into Base 10 and Other Bases

We saw in [Hindu-Arabic Positional System](#) that our Hindu-Arabic system uses **base 10**, which is a system using place values of digits that depend on powers of 10 (or, are based on powers of 10). We've already worked with bases other than base 10: The Babylonian system was base 60, while the Mayan system was base 20.

To explore how our base 10 system is used, answer the following question: What's the following quantity: 4,572? You probably said four thousand five hundred seventy-two (no, there is no "and" between hundred and seventy). But why do you think that 4 means four thousand? A very young person when learning their numbers might say that's a four five seven and two. But you added the context of thousands to the four. Why?

Place value, that's why. You learned early on that where the numeral was gave it different meanings. Ten thousands, thousands, hundreds, tens, and ones. So, you translate that symbol string (4,572) into "four thousand five hundred seventy-two." As we saw in [Hindu-Arabic Positional System](#), expanding a Hindu-Arabic number involved writing the number using each digit times its appropriate power of 10. So, we could write 4,572 as $4 \times 10^3 + 5 \times 10^2 + 7 \times 10^1 + 2 \times 10^0 = 4 \times 1000 + 5 \times 100 + 7 \times 10 + 2 \times 1$.

One possible reason we use base 10 is that we have 10 fingers, and in the cultures where the Hindu-Arabic system developed, that became the standard. Other cultures may have used other ways of organizing numbers, perhaps using 20 by including toes, or using 60 because 60 has many divisors. Mathematically though, base 10 is an awkward base to work in since 10 has limited divisors. But we think it is easy and simple because that's what we've been taught to use.

Using a base 10 system means we need 10 symbols to make our numbering system work: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9.

Now imagine that we all only had 6 fingers instead of 10 and our counting system was based on those 6 fingers. We would be counting in groups of 6, not groups of 10. How would this change how we work with quantity?

First, we'd need only six symbols. Let's use 0, 1, 2, 3, 4, 5. Second, our place values would be based on powers of 6, not powers of 10. For instance, the number 3,024 in base 6 would be $3 \times 6^3 + 0 \times 6^2 + 2 \times 6^1 + 4 \times 6^0$. That is how you can translate a base 6 number into a base 10 number. When we calculate that expression we get $3 \times 6^3 + 0 \times 6^2 + 2 \times 6^1 + 4 \times 6^0 = 3 \times 216 + 0 \times 36 + 2 \times 6 + 4 \times 1 = 648 + 0 + 12 + 4 = 664$.

This means the base 6 number 3,024 is equal to the base 10 number 664.

From now on, if we are using a base 6 number, we will follow it with the subscript 6, like the following: $3,024_6$ means the number is in base 6.

A base 10 number gets no subscript (it's the standard). So, 3,024 is a base 10 number. A base 13 number would be $4,672_{13}$.

So, a base 6 system uses only the symbols 0, 1, 2, 3, 4, and 5. Also, the place values use powers of 6. However, we still don't know how to count in base 6. In order to do so, we'd have to know how to represent the quantities larger than five in base 6. Let's review how our base 10 system works by counting from 0 to 100, which shows how larger values are represented.

In writing the base 10 numbers, you start with these first 10 values:

0	1	2	3	4	5	6	7	8	9
---	---	---	---	---	---	---	---	---	---

But you've run out of symbols. So, we use two digits:

10	11	12	13	14	15	16	17	18	19
----	----	----	----	----	----	----	----	----	----

The 1 out front means you've run out of digits one time.

But now you've run out twice. Continuing with those numbers gives:

20	21	22	23	24	25	26	27	28	29
----	----	----	----	----	----	----	----	----	----

And so on,

30	31	32	33	34	etc....
----	----	----	----	----	---------

Eventually, you hit the 90s,

90	91	92	93	94	95	96	97	98	99
----	----	----	----	----	----	----	----	----	----

And you've run out of the digits again! So, we say we've run out of digits in the tens place one time, hence:

100	101	102	103	104	105	106	107	108	109
-----	-----	-----	-----	-----	-----	-----	-----	-----	-----

That's the pattern we use in base 10. We write out the symbols until we've used all the symbols, then add a digit in front that counts how many times we've used the digits. Knowing the numbers, or being able to count higher and higher, is necessary to understand how all the arithmetic works, as it all goes back to counting.

The counting pattern is the same for any other base, including base 6. So, let's start:

0	1	2	3	4	5
---	---	---	---	---	---

But we've run out of symbols! Just like in base 10, we use a second digit, where the first digit will tell us we've run out of symbols one time.

10	11	12	13	14	15
----	----	----	----	----	----

And we use the same pattern:

20	21	22	23	24	25
30	31	32	33	34	35
40	41	42	43	44	45
50	51	52	53	54	55

But we've run out of symbols for that front digit. So, we indicate it the same way as in base 10...by adding a third digit in front, indicating we've run out of symbols once in the second place:

100	101	102	103	104	105
110	111	etc.			

The symbol pattern is the same, but truncated. We only use the six symbols. So that is how we represent base 6. Being able to write out these numbers is important when working with addition in the base.

When using a base larger than 10, though, we need more symbols. Instead of creating new symbols, we use capital letters, with A representing the digit for "10," B representing the digit for "11," and so on.

EXAMPLE 4.12

Determining Digits of a Base with Less Than 10 Digits

What are the digits used for base 7?

Solution

Since this is base 7, we need only 7 symbols: 0, 1, 2, 3, 4, 5, 6.

YOUR TURN 4.12

1. What are the digits for base 4?

EXAMPLE 4.13

Determining Digits of a Base with More Than 10 Digits

What are the digits used for base 14?

Solution

Since this is base 14, we need 14 symbols. We don't have single character numbers for 10, 11, 12, and 13, so, in a fit of

inspired creativity, we use capital letters A, B, C, D to represent those quantities. So, the digits in base 14 are 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D.

YOUR TURN 4.13

1. What are the digits used for base 12?

WHO KNEW?

Using Base 12

As mentioned in the text, working in base 10 is mathematically awkward. Ten has only two natural number divisors: 2 and 5. This means dividing into groups is not easy. However, 12, or a dozen, has more divisors: 2, 3, 4, and 6. The Dozenal Society recognizes this more mathematically pleasant detail. It advocates for a switch to using base 12 for numbers. Their argument is based on the divisibility of the number 12. But has there ever been a society that used such a system? The answer is yes. A dialect of the Gwandara language in Nigeria uses the base 12 system. It is unlikely, though, that the Dozenal Society will achieve their goal, as the base 10 system is so entrenched in our society.

EXAMPLE 4.14

Converting from One Base into Another

Convert $3,601_7$ into base 10.

Solution

In base 7, the place values are powers of 7. Since there are four digits, the highest power of 7 that is used is 3. This yields $3,601_7 = 3 \times 7^3 + 6 \times 7^2 + 0 \times 7^1 + 1 \times 7^0 = 3 \times 343 + 6 \times 49 + 0 \times 7 + 1 = 1,029 + 294 + 0 + 1 = 1,324$.

VIDEO

[Convert Base 7 to Base 10 \(https://openstax.org/r/Convert_Base_7_to_Base_10\)](https://openstax.org/r/Convert_Base_7_to_Base_10)

YOUR TURN 4.14

1. Convert 421_6 into base 10.

EXAMPLE 4.15

Converting from Base 14 to Base 10

Convert $4B7_{14}$ into base 10.

Solution

In base 14, the place values are powers of 14. Since there are three digits, the highest power of 14 is 2. Also recall that in base 14, 10 is represented by A, 11 is represented by B, 12 is represented by C, and 13 is represented by D. Using that, we convert to base 10:

$$4B7_{14} = 4 \times 14^2 + B \times 14^1 + 7 \times 14^0 = 4 \times 196 + 11 \times 14 + 7 \times 1 = 784 + 154 + 7 = 945.$$

> YOUR TURN 4.15

1. Convert $A3C_{14}$ into base 10.

EXAMPLE 4.16**Converting from Base 12 to Base 10**

Convert $A16_{12}$ into base 10.

✓ Solution

In base 12, the place values are powers of 12. Since there are three digits, the highest power of 12 is 2. Also recall that in base 12, 10 is represented by A and 11 is represented by B. Using that, we convert to base 10:

$$A16_{12} = 10 \times 12^2 + 1 \times 12^1 + 6 \times 12^0 = 10 \times 144 + 1 \times 12 + 6 \times 1 = 1,440 + 12 + 6 = 1,458.$$

> YOUR TURN 4.16

1. Convert $5AB_{12}$ into base 10.

EXAMPLE 4.17**Converting from Base 2 to Base 10**

Convert 1011_2 into base 10.

✓ Solution

In base 2, the place values are powers of 2. Since there are four digits, the highest power of 2 is 3. Using that, we convert to base 10:

$$1011_2 = 1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 1 \times 2^0 = 1 \times 8 + 0 \times 4 + 1 \times 2 + 1 \times 1 = 8 + 0 + 2 + 1 = 11.$$

> YOUR TURN 4.17

1. Convert 11011_2 into base 10.

? WHO KNEW?**Before Napoleon**

Before Napoleon's France, which adopted the base 10 system, a modified base 12 system was often used in Europe. Twelve is easily divisible into groups of 2, 3, 4, and 6, which makes it easier to work with. Even our numbering system retains a bit of this. You have likely noticed that we use the words *thirteen*, *fourteen*, *fifteen*, and so on to indicate 10 and 3, 10 and 4, 10 and 5, and so on. Even the 20s reinforce this idea, as in twenty-one, and twenty-two. However, two numbers don't follow this pattern, namely 11 and 12. If they followed the same rules, they'd be one teen and two teen. We even have a special word for 12; that is, a dozen. However, etymologically speaking, the words *eleven* and *twelve* are likely derived by referencing the number 10. These two numbers may date back to the Old English words *endleofan* and *twelf*, which can be traced back further to *ain lif* and *twa lif*. The word *lif* here may be the base word for "to leave." This would suggest *ain lif* is one left after 10, and *twa lif* is two left after 10, or, 11 and 12.

EXAMPLE 4.18**Writing Numbers in Base Systems Other Than Base 10**

Write the numbers in base 7 up to 100_7 .

✓ **Solution**

Step 1: Using the patterns we indicated earlier, we begin with the first seven digits.

0, 1, 2, 3, 4, 5, 6

Step 2: Since we've run out of digits, we start with 10, indicating we've run out of symbols once.

10, 11, 12, 13, 14, 15, 16

Step 3: Continuing in the same way, we get:

20, 21, 22, 23, 24, 25, 26

30, 31, 32, 33, 34, 35, 36

40, 41, 42, 43, 44, 45, 46

50, 51, 52, 53, 54, 55, 56

60, 61, 62, 63, 64, 65, 66

Now, all the digits have been used in the leading digits. Since the digits have all been used in that leading digit, we use 100, as in base 10.

100

> YOUR TURN 4.18

1. Write the numbers of base 4 up to 100_4 .

EXAMPLE 4.19**Writing Numbers in Bases with More Than 10 Symbols**

Write the numbers in base 14 up to 100_{14} .

✓ **Solution**

Step 1: Using the patterns we indicated earlier, we begin with the first 14 digits.

0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D

Step 2: Since we've run out of digits, we start with 10, indicating we've run out of symbols once.

10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 1A, 1B, 1C, 1D

Step 3: Continuing in the same way, we get:

20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 2A, 2B, 2C, 2D

30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 3A, 3B, 3C, 3D

40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 4A, 4B, 4C, 4D

50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 5A, 5B, 5C, 5D

60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 6A, 6B, 6C, 6D

70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 7A, 7B, 7C, 7D

80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 8A, 8B, 8C, 8D

90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 9A, 9B, 9C, 9D

A0, A1, A2, A3, A4, A5, A6, A7, A8, A9, AA, AB, AC, AD
 B0, B1, B2, B3, B4, B5, B6, B7, B8, B9, BA, BB, BC, BD
 C0, C1, C2, C3, C4, C5, C6, C7, C8, C9, CA, CB, CC, CD
 D0, D1, D2, D3, D4, D5, D6, D7, D8, D9, DA, DB, DC, DD
 100

YOUR TURN 4.19

1. Write the numbers of base 12 up to 100_4 .

Base 2 is important in the digital age, as it is the system used by computers. It is the simplest base to work with, but has the drawback that the numbers in base 2 may use many, many digits. In [Addition and Subtraction in Base Systems](#) and [Multiplication and Division in Base Systems](#), we will look at base 2 in each situation.

EXAMPLE 4.20

Writing Numbers in Base 2

Write the numbers in base 2 up to 100_2 .

Solution

Base 2 uses only two symbols: 0 and 1. Following the pattern established previously, the numbers in base 2 up to 100_2 are 0, 1, 10, 11, and 100.

YOUR TURN 4.20

1. Write the numbers in base 3 up to 100_3 .

WHO KNEW?

Early Hawaiian Numeration System

Before the British arrived in Hawaii, people there used a system that combined two different bases. Objects were initially grouped into collections of four, and a collection of four was referred to as *kauna*. A person could have two *kauna* and three “ones” (in Hindu-Arabic, 11). Or they could have eight *kauna* and one “ones” (In Hindu-Arabic, 33). However, sets of four were grouped in collections of 10. A set of 10 *kauna* was *ka’au*. The collections of *ka’au* were grouped by 10 also. Which meant that 10 *ka’au* (this is 40 in Hindu-Arabic) would be *lau* (or 400 in Hindu-Arabic). What this shows is that the Hawaiian culture developed a system that used base 4 combined with base 10.

Conversion of Base 10 into Another Base

Converting from base 10 into another base uses repeated division, recording the **remainder** at each step. Then, the number in the new base is the remainder starting from the last remainder found. To be accurate in what we’re saying, we need to remind ourselves of some terminology associated with division. When integers are divided, the one being divided is the **dividend**, and the one that is dividing the dividend is the **divisor**. The **quotient** is the largest natural number that can be multiplied by the divisor where the product is less than the dividend.

When the integer n is divided by the integer d , n is called the **dividend** and d is the **divisor**.

To convert a base 10 number n into base d , we divide n by d , recording the remainder. Then we divide the quotient from that step by the base d , and record the remainder again. We continue this process until the quotient is 0. Then, the base d number has digits that start with the last remainder and use each remainder in reverse order.

EXAMPLE 4.21**Converting from Base 10 into a Lower Base**

Convert 298 to base 6.

 Solution

We divide 298 by 6, and record the remainder. Then we divide the quotient from that step by 6, and record the remainder again. We continue this process until the quotient is 0. Then, the base 6 number has digits that start with the last remainder and use each remainder in reverse order.

Step 1: When we divide 298 by 6, we get $\begin{array}{r} 49r4 \\ 6 \overline{)298} \end{array}$. The quotient is 49 and the remainder is 4.

Step 2: Now we divide the quotient, 49, by 6. This gives $\begin{array}{r} 8r1 \\ 6 \overline{)49} \end{array}$. The quotient is 8 and the remainder is 1.

Step 3: Repeating, we get $\begin{array}{r} 1r2 \\ 6 \overline{)8} \end{array}$. The quotient is 1 and the remainder is 2.

Step 4: Finally, we perform the operation on the quotient 1, $\begin{array}{r} 0r1 \\ 6 \overline{)1} \end{array}$ giving us a quotient of 0 and a remainder of 1.

Step 5: The base 6 number has digits equal to the remainders in reverse order, 1214₆. So, 298 in base 10 when converted to base 6 is 1214₆.

 YOUR TURN 4.21

1. Convert 693 to base 7.

 VIDEO

[Converting from Base 10 to Another Base \(https://openstax.org/r/Base_10_to_Another_Base\)](https://openstax.org/r/Base_10_to_Another_Base)

EXAMPLE 4.22**Converting from Base 10 into a Higher Base**

Convert 45,134 to base 13.

 Solution

We divide 45,134 by 13, and record the remainder. Then we divide the quotient from that step by 13, and record the remainder again. We continue this process until the quotient is 0. Then, the base 13 number has digits that start with the last remainder and use each remainder in reverse order. It is at this step that we'll convert to the base 13 digits, which are 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C.

Step 1: When we divide 45,134 by 13, we get $\begin{array}{r} 3,471r11 \\ 13 \overline{)45,134} \end{array}$. The quotient is 3,471 and the remainder is 11.

Step 2: Now we divide the quotient, 3,471, by 13. This gives $\begin{array}{r} 267r0 \\ 13 \overline{)3,471} \end{array}$. The quotient is 267 and the remainder is 0.

Step 3: Repeating, we get $\begin{array}{r} 20r7 \\ 13 \overline{)267} \end{array}$. The quotient is 20 and the remainder is 7.

Step 4: Again, and we get $\begin{array}{r} 1r7 \\ 13 \overline{)20} \end{array}$. The quotient is 1 and the remainder is 7.

Step 5: Finally, we get $\begin{array}{r} 0r1 \\ 13 \overline{)1} \end{array}$, with quotient 0 and a remainder 1.

Step 6: The base 13 number has digits equal to the remainders in reverse order, which were 1, 7, 7, 0, and 11. The 11 is written as B in base 13. So, 45,134 in base 10 when converted to base 13 is $1770B_{13}$.

> YOUR TURN 4.22

1. Convert 9,275 to base 12.

EXAMPLE 4.23

Converting from Base 10 into Base 2

Convert 100 to base 2.

✓ Solution

Following the pattern above:

Step 1: We divide 100 by 2, and record the remainder.

Step 2: Then we divide the quotient from that step by 2, and record the remainder again.

Step 3: We continue this process until the quotient is 0.

Step 4: Following this process, the remainders are, in order, 0, 0, 1, 0, 0, 1, 1. Writing those in reverse order gives the number in base 2, 1100100_2 .

Notice that 100 in base 2 took seven digits.

> YOUR TURN 4.23

1. Convert 137 to base 2.

Converting from Hindu-Arabic Numbers to Mayan Numbers

To convert from a Hindu-Arabic number to a Mayan number involves two distinct processes. First, the number must be converted to base 20, using the process described and demonstrated previously. Next, that base 20 number has to be written using Mayan numerals. For reference, the Mayan numerals and their values are below.

0	1	2	3	4
	•	••	•••	••••
5	6	7	8	9
10	11	12	13	14
15	16	17	18	19

EXAMPLE 4.24**Converting from Base 10 into the Mayan System**

Convert the following into Mayan numbers.

- 51
- 653

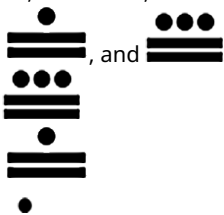
✓ **Solution**

- The Mayan system is base 20, so we must use 20 in the process from above. The first division has a quotient of 2 and remainder of 11. The 11 serves as the “ones” digit. Dividing that quotient, 2, by 20 has a quotient of 0 with a remainder of 2. The 2 becomes the “twenties” digit of the number. So, in base 20, the number would be 2 followed

by 11. The Mayan symbols for 2 and 11 are  and . Writing these vertically, with the “ones” digit on top, as appropriate for Mayan numbers, results in:



- The Mayan system is base 20, so we must use 20 in the process from above. The first division, 673 divided by 20, has a quotient of 32 and remainder of 13. Dividing that quotient, 32, by 20 has a quotient of 1 with a remainder of 12. Dividing that quotient, 1, by 20 has a quotient of 0 and a remainder of 1. Since there are three remainders here, this is a three-digit number. The 1 is the “20-squared” digit, the 12 is the “twenties” digit, and the 13 is the “ones” digit. So, in base 20, the number would be 1 followed by 12 followed by 13. The Mayan symbols for 1, 12 and 13 are ,



, and . Writing these vertically, as appropriate for Mayan numbers, would result in:

> YOUR TURN 4.24

Convert the following into Mayan numbers.

- 137
- 2,171

? WHO KNEW?**Other Languages, Other Bases**

There have been base systems that use bases other than 10. Some bases used were 20, 12, and 27! Visit [this site to see more on the languages that used other bases \(https://openstax.org/r/number-systems_other_languages\)](https://openstax.org/r/number-systems_other_languages).

Errors in Converting Between Bases

There are some common errors that are made when converting between bases. Often, it comes down to using an “illegal” symbol in the new base.

EXAMPLE 4.25**Detecting an Illegal Symbol When Converting Between Bases**

A base 10 number is converted to base 7 and the result was 2081₇. Was an error committed? How do you know?

 Solution

The result has the digit 8 in it. In base 7, 8 is an illegal symbol. Based on that, an error was committed.

 YOUR TURN 4.25

1. A base 10 number is converted to base 4 and the result was 3702_4 . Was an error committed? How do you know?

When converting from base 10 to another base, an illegal symbol will be used if a mistake was made in the division process used to find the number in the new base. Since the digits are based on the remainders, any remainder that is an illegal symbol would indicate an error.

EXAMPLE 4.26**Detecting an Error in Division When Converting Between Bases**

When changing from base 10 to base 8, the division process resulted in the following remainders: 1, 0, 9, 2, 4. Was an error committed? How do you know?

 Solution

The remainders include 9, which in base 8 is an illegal symbol.

 YOUR TURN 4.26

1. When changing from base 10 to base 6, the division process resulted in the following remainders: 5, 0, 0, 10. Was an error committed? How do you know?

Another possible way to detect an error in converting between bases is to count the number of digits. When converting from a higher base to a lower base, the number of digits cannot get smaller. Similarly, when converting from a lower base to a higher base, the number of digits cannot get bigger. So, if a base 10 number is converted to a base 3 number, the number of digits in the new base 3 number cannot be less than the number of digits in the base 10 number. Similarly, if a base 7 number is converted to base 10, the number of digits in the base 10 number cannot be more than the number of digits in the original base 7 number.

EXAMPLE 4.27**Detecting an Error in Number of Digits When Converting Between Bases**

A five-digit base 10 number is converted to a base 5 number. The base 5 number has four digits. Was an error committed? How do you know?

 Solution

Since 10 is larger than 5, the base 5 number cannot have less digits than the base 10 number. Since it did, we know an error has been made.

 YOUR TURN 4.27

1. A six-digit base 12 number is converted into a base 10 number. The base 10 number has five digits. Was an error committed? How do you know?

Check Your Understanding

14. In base 25, how many symbols would be necessary?
15. In base 18, what would the place value of the 4 be in the number 348_{18} be?

16. Convert 2304_5 into base 10.
17. When counting in base 9, what number would follow 38_9 ?
18. Convert 329 into base 8.
19. Convert ABC_{14} into base 10.
20. How do you know a mistake was made when converting from base 10 to base 4 and the result is 152_4 ?



SECTION 4.3 EXERCISES

1. What does it mean when we say a number is written in base 7?
2. What does it mean when we say a number is written in base 12?
3. How many symbols are there in a base 3 system? What are they?
4. How many symbols are there in a base 15 system? What are they?
5. List the numbers, up to 100, in the base 5 system.
6. List the numbers, up to 100, in the base 2 system.

For the following exercises, convert the number into a base 10 number.

7. 14_5
8. 21_6
9. 12_3
10. 34_5
11. 14_8
12. 78_9
13. $3B_{12}$
14. 241_6
15. 101_2
16. $4A7_{14}$
17. 804_9
18. 101001_2
19. 3223_6
20. 1436_7
21. $8A0BD_{15}$
22. 110202_3
23. $100A4_{12}$
24. 1010000101_2

For the following exercises, convert the base 10 number into the given base.

25. 12 into base 4
26. 25 into base 2
27. 43 into base 12
28. 153 into base 5
29. 203 into base 2
30. 431 into base 4
31. 543 into base 12
32. 1,023 into base 2
33. 2,876 into base 4
34. 1,765 into base 5
35. 1,993 into base 7
36. 2,000 into base 2
37. 4,368 into base 12
38. 12,562 into base 16 (Hint: Base 16 uses the symbols 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E, F.)
39. When converting from base 10 to base 6, a student finds the following remainders: 3, 1, 3, 6. How do you know that a mistake was made?
40. When converting from base 10 to base 4, a student finds the following remainders: 0, 0, 3, 7, 2. How do you know a